Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

MERN

Practical II

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Class: TYIT	Batch: 1
Date of Assignment: 30-6-26	Date/Time of Submission: 30-6-26

2. Core JavaScript Concepts for MERN

a. Create functions and call them with parameters.

CODE

```
function calculateSalary(name, basicSalary) {
    let bonus = 5000;
    let totalSalary = basicSalary + bonus;

    console.log("Employee Name:", name);
    console.log("Basic Salary:", basicSalary);
    console.log("Bonus:", bonus);
    console.log("Total Salary:", totalSalary);
}

calculateSalary("Amit", 30000);
console.log("Sneha");
```

OUTPUT

```
PS C:\MERN Practical> node pract2a.js
Employee Name: Amit
Basic Salary: 30000
Bonus: 5000
Total Salary: 35000
Sneha
```

CODE

```
function genbill(product, quantity, price) {
    let total = quantity * price;
    let totalBill = total + (total*0.18);
    console.log("Product Name:", product);
    console.log("Quantity:", quantity);
```

```
    console.log("Price per Item:", price);
    console.log("Total Bill:", totalBill);

}

genbill("Mobile", 2,8000);
console.log("Sneha");
```

OUTPUT

```
PS C:\MERN Practical>  node pract2a.js
Product Name: Mobile
Quantity: 2
Price per Item: 8000
Total Bill: 18880
Sneha
```

b. Work with arrays and objects in JavaScript.

CODE

```
let fruits = ["Mango", "Apple", "Pear"];

let Student = {
    name: "Sneha",
    age: 19,
    city: "Mumbai"
};

console.log("Fruits:", fruits);
console.log("Second Fruit:", fruits[1]);
console.log("Student Name:", Student.name);
console.log("Student Age:", Student.age);
console.log("Student City:", Student.city);
```

OUTPUT

```
PS C:\MERN Practical>  node pract2b.js
Fruits: [ 'Mango', 'Apple', 'Pear' ]
Second Fruit: Apple
Student Name: Sneha
Student Age: 19
Student City: Mumbai
```

c. Demonstrate use of arrow functions.

CODE

```
let add = (a, b) => {
```

```
    return a + b;
};

console.log("SUM =", add(20, 10));

let sub = (a, b) => {
    return a - b;
};

console.log("SUB =", sub(20, 10));

let mul = (a, b) => {
    return a * b;
};

console.log("MUL =", mul(20, 10));
console.log("Sneha");
```

OUTPUT

```
PS C:\MERN Practical>  node pract2c.js
SUM = 30
SUB = 10
MUL = 200
Sneha
```

d. Write a program using array methods (map, filter, reduce).

CODE

```
let numbers = [10,20,30,40,50];
let doubled = numbers.map(num => num*2);
let greater25 = numbers.filter(num => num >25);
let total = numbers.reduce((sum,num) => sum + num,0);

console.log("Original Array: ",numbers);
console.log("Map: ",doubled);
console.log("Filter: ",greater25);
console.log("Reduce: ",total);
console.log("Sneha");
```

OUTPUT

```
PS C:\MERN Practical> node pract2d.js
Original Array:  [ 10, 20, 30, 40, 50 ]
Map:  [ 20, 40, 60, 80, 100 ]
Filter:  [ 30, 40, 50 ]
Reduce:  150
Sneha
```

e. Create a program that manipulates objects and displays output.

CODE

```
let student = {
    rollno: 27,
    name:  "Sneha",
    marks:  90
};
console.log("Original Object: ");
console.log(student);

console.log("\n Accessing Properties");
console.log("Name: ",student.name);
console.log("Marks: ", student.marks);

student.marks = 95;
console.log("\n After Updating Marks");
console.log(student);

student.city = "Mumbai";
console.log("\n After Adding City");
console.log(student);
console.log("Sneha");
```

OUTPUT

```
PS C:\MERN Practical> node pract2e.js
Original Object:
{ rollno: 27, name: 'Sneha', marks: 90 }

 Accessing Properties
Name:  Sneha
Marks:  90

 After Updating Marks
{ rollno: 27, name: 'Sneha', marks: 95 }

 After Adding City
{ rollno: 27, name: 'Sneha', marks: 95, city: 'Mumbai' }
Sneha
```

f. Implement a small JavaScript program combining functions, arrays, and objects.

CODE

```
let students = [
    {
        rollNo: 101 ,
        name: "Rahul",
        marks: 85
    },
    {
        rollNo: 102 ,
        name: "Varun",
        marks: 90
    },
    {
        rollNo: 103 ,
        name: "Amit",
        marks: 95
    }
];
function display(s)
{
    console.log("Student Details");
    console.log("---------------------------");
    for (let stud of s)
    {
        console.log(`Roll No : ${stud.rollNo}`);
```

```
        console.log(`Name : ${stud.name}`);
        console.log(`Marks: ${stud.marks}`);
        console.log("---------------------------");
    }
}
display(students);
console.log("Sneha");
```

OUTPUT

```
PS C:\MERN Practical> node pract2f.js
Student Details
---------------------------
Roll No : 101
Name : Rahul
Marks: 85
---------------------------
Roll No : 102
Name : Varun
Marks: 90
---------------------------
Roll No : 103
Name : Amit
Marks: 95
---------------------------
Sneha
```